

DESERTO DE ATACMA, Chile

WMO: 854670

Lat: 27.264S

Lon: 70.774W

Elev: 689

StdP: 14.33

Time Zone: -4.00 (CHL)

Period: 05-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	42.6	44.5	35.2	30.8	49.7	37.8	34.2	49.3	20.0	57.9	18.1	58.0	5.3	100	0.461

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	13.9	77.2	63.4	75.4	63.0	73.7	62.2	64.8	74.7	63.8	73.4	62.9	72.3	11.4	260

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
60.8	81.7	69.0	59.3	77.4	67.6	58.9	76.2	67.2	30.2	74.8	29.4	74.0	28.8	72.6	70.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
18.4	16.4	14.7	DB	38.4	82.3	1.5	2.9	37.4	84.3	36.5	86.0	35.6	87.6	34.5	89.7	
			WB	35.9	66.5	1.8	1.8	34.6	67.8	33.6	68.8	32.6	69.8	31.3	71.1	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	60.3	67.4	67.7	65.2	61.1	58.1	55.2	54.2	55.2	56.3	58.5	61.2	64.0
	DBStd	5.25	2.40	2.30	2.55	2.75	2.73	2.73	2.59	2.66	2.48	2.16	2.06	2.24
	HDD50	4	0	0	0	0	0	1	2	1	0	0	0	0
	HDD65	1928	5	3	28	120	214	294	336	305	260	202	115	46
	CDD50	3764	538	496	472	334	252	157	131	161	190	263	336	434
	CDD65	213	79	78	35	4	1	0	0	0	0	0	1	15
	CDH74	355	141	146	42	8	3	1	1	6	0	0	1	6
	CDH80	13	5	7	0	1	0	0	0	0	0	0	0	0
Wind	WSAvg	7.4	7.7	7.7	7.3	6.9	6.3	6.5	6.8	7.3	7.5	8.2	8.3	8.2
Precipitation	PrecAvg	1.0	0.0	0.0	0.0	0.0	0.1	0.4	0.2	0.2	0.0	0.0	0.0	0.0
	PrecMax	4.8	0.0	0.0	0.2	0.6	1.0	2.4	0.8	2.4	0.0	0.1	0.0	0.1
	PrecMin	0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
	PrecStd	1.3	0.0	0.0	0.0	0.2	0.2	0.7	0.2	0.6	0.0	0.0	0.0	0.0
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	80.6	80.7	77.3	74.8	72.0	70.0	68.3	71.7	69.7	70.0	72.0	75.0
		MCWB	65.6	65.6	63.5	61.4	58.4	54.7	54.6	54.0	57.3	58.2	59.4	61.9
	2%	DB	77.2	77.3	75.1	71.6	68.2	65.9	64.7	65.8	66.1	67.9	70.0	73.2
		MCWB	63.7	63.9	62.8	59.9	57.9	54.9	54.7	54.3	56.1	57.2	58.9	60.8
	5%	DB	75.3	75.4	73.2	69.7	66.1	62.9	62.5	62.9	64.2	66.0	68.3	71.6
		MCWB	63.2	63.4	62.0	59.5	56.9	54.2	53.8	53.7	54.8	56.3	58.1	60.1
	10%	DB	73.6	73.8	71.4	67.6	64.1	61.0	60.4	60.9	62.3	64.2	66.6	69.9
		MCWB	62.5	62.7	61.3	58.7	56.0	53.9	53.2	53.5	53.9	55.3	57.3	59.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	67.2	66.4	65.1	62.3	60.8	57.5	57.4	57.0	58.1	58.6	60.3	62.7
		MCDB	76.3	78.0	74.8	71.5	69.5	65.8	64.7	67.5	67.5	68.1	70.0	72.7
	2%	WB	65.1	64.8	63.4	60.9	58.8	56.2	55.7	55.6	56.6	57.5	59.3	61.4
		MCDB	74.8	75.1	72.5	69.2	66.3	63.3	62.6	63.3	64.8	66.3	68.6	70.8
	5%	WB	63.8	64.0	62.4	59.8	57.5	55.1	54.6	54.5	55.5	56.6	58.3	60.7
		MCDB	73.6	73.7	71.2	67.4	64.1	61.4	60.9	61.4	62.5	64.7	67.0	69.6
	10%	WB	62.7	63.0	61.6	58.6	56.6	54.0	53.3	53.5	54.4	55.8	57.5	60.0
		MCDB	72.5	72.4	69.6	65.6	62.5	59.8	59.0	59.7	60.7	63.0	65.5	68.4
Mean Daily Temperature Range	5% DB	MDBR	13.2	13.9	13.0	13.7	13.1	14.2	14.4	13.9	13.7	13.0	12.8	13.0
		MCDBR	14.9	15.3	15.0	16.6	16.6	17.5	18.2	17.6	16.3	15.5	14.4	14.5
		MCWBR	5.8	6.2	6.3	7.9	8.8	9.7	10.1	9.2	8.3	7.5	6.4	6.0
	5% WB	MCDBR	13.9	14.2	14.0	15.2	15.3	15.7	16.6	15.8	15.0	14.6	13.6	13.5
		MCWBR	5.6	6.0	6.1	7.4	8.4	9.2	9.6	8.6	7.8	7.2	6.2	5.8
Clear-Sky Solar Irradiance	taub	0.369	0.362	0.349	0.345	0.322	0.308	0.304	0.310	0.327	0.325	0.330	0.345	
	taud	2.378	2.399	2.428	2.415	2.469	2.506	2.502	2.481	2.418	2.440	2.446	2.419	
	Ebn at Noon	309	305	300	285	278	276	282	293	302	314	319	317	
	Edh at Noon	41	39	36	33	29	26	27	31	36	37	38	40	
All-Sky Solar Radiation	RadAvg	2470	2321	1918	1471	1109	1008	1101	1356	1739	2072	2333	2466	
	RadStd	99	90	82	80	76	54	46	70	62	91	87	129	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page